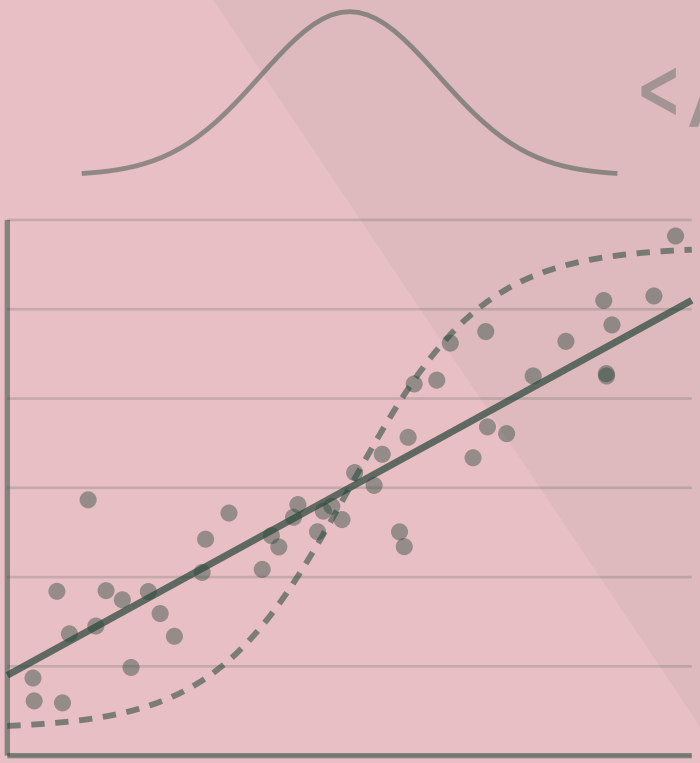
VENVLEARNING

STUDENT WORKBOOK



```
proc means;    df.describe()    summary(x)
```

Data Science & Psychometrics

YEAR 1

Practice, coding exercises, weekly quizzes, and answers

2026–2027 Professional Learning · 2 units

A VenvLearning professional learning curriculum

SAS · PYTHON · R

VENVLEARNING

Professional Learning
Data Science & Psychometrics

STUDENT WORKBOOK

2026–2027 Professional Learning

Unit 1 Your Data Toolkit

Unit 2 Data Wrangling

This book belongs to _____

A VenvLearning professional learning curriculum

How to use this workbook

Every lesson in your textbook has matching practice here: a warm-up, practice problems, a coding exercise, and an exit question. Fridays have a 15-minute quiz; the last Friday of each unit is the unit exam.

- Work a problem before checking the Answer Key at the back.
- For coding exercises, write your code in SAS Studio, RStudio, or Jupyter, then record the key output here.
- If an answer doesn't match, compare your code with the solution line by line. Finding the difference is the fastest way to learn.
- Use the Apply It prompts on Fridays to try the week's skill on your own work data.

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UNIT 1

Your Data Toolkit

The analysis workflow; reading data in SAS, Python, and R; types and levels of measurement; reproducible habits.

Your practice pages for Unit 1. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

workflow	Steps from question to communication.
data frame	A table of rows and columns.
ordinal	Ordered categories.
reproducible	Can be rerun with the same result.

LESSON 1.1 · PRACTICE
Planned for Mon Nov 2

Unit 1: Your Data Toolkit

Lesson 1.1: Your Toolkit and the Analysis Workflow

Name _____ Date _____

TODAY'S GOAL

I can describe the steps of a data analysis and set up SAS, Python, and R so I can practice every day.

Standards: Foundations · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 1.1 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Set Up, Part C Reflect.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1. What does IDE stand for?

2. Why does reproducible work matter?

PART B Set Up

3. Check off each step and write where you'll work: (1) SAS Studio opened; (2) R and RStudio installed or Posit Cloud account; (3) Python with Jupyter ready; (4) data pack downloaded into a folder named data.

4 Run the Hello, data example in all three languages. Record what each one printed.

PART C Reflect

5 Which workflow step do you feel most and least confident about at work? Why?

6 Write the recurring work task you'll rebuild in code this year.

LESSON 1.2 · PRACTICE
Planned for Tue Nov 3

Unit 1: Your Data Toolkit

Lesson 1.2: Reading Data into SAS, Python, and R

Name _____ Date _____

TODAY'S GOAL

I can import a CSV file in all three languages and check its size and first rows.

Standards: Import · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 1.2 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 CSV stands for…

2 SAS's name for a data frame

PART B Practice

3 Import examinees.csv in the language of your choice. How many rows and columns does it have?

4 What are the gender, region, and program of examinee E1002?

5 How many blank (omitted) responses does exam.csv have on item i01?

6 Write the code in a second language to print the first 3 rows of examinees.

PART C Exit Ticket

7 List the three checks you should do after every import.

LESSON 1.3 · PRACTICE
Planned for Wed Nov 4

Unit 1: Your Data Toolkit

Lesson 1.3: Variables, Data Types, and Levels of Measurement

Name _____ Date _____

TODAY'S GOAL

I can identify data types and levels of measurement and explain why they decide which analyses make sense.

Standards: Foundations · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 1.3 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Classify, Part C Code It, Part D Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 Categories with no order

2 Equal spacing and a true zero

PART B Classify

3 prior_gpa

4 left_next_year (0/1)

5 tenure_years

6 dept

7 intent_stay (1–10 rating)

8 rank in class

PART C Code It

9 Load survey.csv and show each column's type. Which columns are numbers but really ordinal?

PART D Exit Ticket

10 Is a mean of 3.02 on one Likert item meaningful? Give one argument each way.

LESSON 1.4 · PRACTICE
Planned for Thu Nov 5

Unit 1: Your Data Toolkit

Lesson 1.4: Reproducible Work Habits

Name _____ Date _____

TODAY'S GOAL

I can organize a project, comment code, read a log, and save outputs so an analysis can be rerun.

Standards: Foundations · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 1.4 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 Folder for the original data

2 A system that records every change to files

PART B Practice

3 Log line “WARNING: Variable X is uninitialized.” What should you do?

4 Better file name: final_v2_NEW.sas or 2026-11-05_scoring.sas?

5 Why rerun a script from the top before sharing?

6 Write code to save only the Evening examinees to evening_examinees.csv. How many rows?

PART C Exit Ticket

7 Describe the folder structure you created for your work project.

UNIT REVIEW
Planned for Thu Nov 5 (homework) or as needed

About 30 minutes

Unit 1 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Review

1. Six workflow steps

2. Level of measurement for program

3. Three checks after import

4. Where to fix a data error

WEEKLY QUIZ
Planned for Fri Nov 6

About 15 minutes

Week 1 Quiz

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 10 points.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A Concepts (1 point each)

1. Which comes right after importing data in the workflow?

   A modeling
   B cleaning and tidying
   C communicating

2. A 1–5 agreement item is measured at which level?

   A nominal
   B ordinal
   C ratio

3. Hours of study is measured at which level?

   A ordinal
   B interval
   C ratio

4. In tidy data, what does each row represent?

   _____

5. What should you never edit by hand?

   _____

PART B Read the code (1 point each)

6 What does this print?

```
x <- 17.67 * 2
x
```

7 In pandas, what does exam.shape return for exam.csv?

8 What does proc contents show?

PART C Apply It (2 points)

9 Name one recurring task at work, the level of measurement of its main variable, and one summary that fits it.

UNIT 2

Data Wrangling

Filtering, scoring, grouped summaries, missing data, joins, reshaping, cleaning, validation, SQL, functions, dates, text, and codebooks.

Your practice pages for Unit 2. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

filter	Keep rows that meet a condition.
left join	Keep all rows of the main table.
long format	One row per person-item.
codebook	Documentation of every variable.
subscore	A score on part of a test.
validation check	A rule the data must satisfy.

LESSON 2.1 · PRACTICE
Planned for Mon Nov 9

Unit 2: Data Wrangling

Lesson 2.1: Selecting Columns and Filtering Rows

Name _____ Date _____

TODAY'S GOAL

I can keep the columns I need and the rows that meet one or more conditions.

Standards: Wrangling · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.1 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 Keep only certain columns =

2 SAS statement to keep rows

PART B Practice

3 How many examinees are in the South region and the Evening program?

4 How many examinees have prior_gpa of 3.5 or higher?

5 Keep Day or Online examinees (use “in a list”). How many rows?

PART C Exit Ticket

6 Why check the row count after every filter?

LESSON 2.2 · PRACTICE
Planned for Tue Nov 10

Unit 2: Data Wrangling

Lesson 2.2: Creating Variables: Scoring a Test

Name _____ Date _____

TODAY'S GOAL

I can create new variables, including scoring multiple-choice responses against a key and computing a total score.

Standards: Wrangling · Psychometrics · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.2 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 1 for a match, 0 otherwise is called...

2 Key for item i01

PART B Practice

3 What proportion of examinees answered i01 correctly?

4 What proportion answered i05 correctly?

5 Score the test and report the mean, minimum, and maximum total.

6 Create pass = 1 if total >= 20, else 0. How many passed?

PART C Exit Ticket

7 Explain in plain words what the scoring code does.

LESSON 2.3 · PRACTICE
Planned for Wed Nov 11

Unit 2: Data Wrangling

Lesson 2.3: Sorting, Counting, and Grouped Summaries

Name _____ Date _____

TODAY'S GOAL

I can sort data, count categories, and summarize a variable for each group.

Standards: Wrangling · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.3 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 Count of rows in each category

2 SAS procedure for counts

PART B Practice

3 How many Day examinees are female (F)?

4 Count examinees by region. Which region is largest?

5 Sort examinees by hours_prep, largest first. What is the maximum?

PART C Exit Ticket

6 Describe split–apply–combine in your own words with a work example.

LESSON 2.4 · PRACTICE
Planned for Thu Nov 12

Unit 2: Data Wrangling

Lesson 2.4: Missing Data: Finding and Handling It

Name _____ Date _____

TODAY'S GOAL

I can count missing values, describe why they matter, and choose a reasonable first strategy.

Standards: Wrangling · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.4 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 pandas: count missing per column

2 R: rows with no missing values

PART B Practice

3 How many respondents are missing at least one item?

4 Mean of e1 using all available responses

5 Compute each person's mean on e1–e4, using answered items only. Then count people who answered fewer than 3 of the 4.

PART C Exit Ticket

6 Why can 3% missing per item turn into 33% incomplete rows?

WEEKLY QUIZ
Planned for Fri Nov 13

About 15 minutes

Week 2 Quiz

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Concepts and code (1 point each)

1. Which keeps rows?

   A select
   B filter
   C mutate

2. pandas operator for “and” between conditions

3. What does this return? `(examinees["program"] == "Online").sum()`

4. How is an omitted response scored in the Unit 2 scoring code?

5. E1002's total score

6. Which SAS procedure counts categories?

   A PROC MEANS
   B PROC FREQ
   C PROC SORT

7. How many survey respondents have complete data on all 12 items?

8 Why might listwise deletion bias a survey result?

LESSON 2.5 · PRACTICE
Planned for Mon Nov 16

Unit 2: Data Wrangling

Lesson 2.5: Joining Tables

Name _____ Date _____

TODAY'S GOAL

I can join two tables on a key and check that the join kept the rows I expected.

Standards: Wrangling · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.5 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 The column used to match rows

2 Join that keeps every row of the main table

PART B Practice

3 Mean total for females (F)

4 Mean total for the South region

5 What is the correlation between hours_prep and total? (Join first.)

6 What would you suspect if a left join turned 600 rows into 612?

PART C Exit Ticket

7 Write the question you'd ask before choosing left vs inner join.

LESSON 2.6 · PRACTICE
Planned for Tue Nov 17

Unit 2: Data Wrangling

Lesson 2.6: Reshaping Data: Wide and Long

Name _____ Date _____

TODAY'S GOAL

I can reshape item data from wide (one column per item) to long (one row per person-item) and back.

Standards: Wrangling · Psychometrics · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.6 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 One row per person, one column per item

2 R function for wide → long

PART B Practice

3 Which three items are easiest?

4 Add each item's domain to the long data (join with exam_key) and compute mean proportion correct by domain.

PART C Exit Ticket

5 Give one analysis that's easier in wide format and one that's easier in long.

LESSON 2.7 · PRACTICE
Planned for Wed Nov 18

Unit 2: Data Wrangling

Lesson 2.7: Cleaning Messy Data

Name _____ Date _____

TODAY'S GOAL

I can standardize text categories, convert numbers stored as text, and parse dates in mixed formats.

Standards: Wrangling · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.7 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 Remove spaces at the ends of text

2 Convert, turning failures into missing

PART B Practice

3 How many distinct raw spellings does program have?

4 Status counts after cleaning

5 What is the mean cleaned fee (all 63 rows, ignoring missing)?

6 In which month did most enrollments occur?

PART C Exit Ticket

7 Why fix data in code rather than in Excel?

LESSON 2.8 · PRACTICE
Planned for Thu Nov 19

Unit 2: Data Wrangling

Lesson 2.8: Duplicates and Validation Checks

Name _____ Date _____

TODAY'S GOAL

I can find and remove duplicate records and write checks that catch impossible values.

Standards: Wrangling · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.8 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 pandas: remove exact duplicates

2 R: stop if a condition fails

PART B Practice

3 After de-duplicating, how many Active enrollments?

4 Withdrawn by program (Day, Evening, Online)

5 Write a validation check that every total score in your scored data is between 0 and 30.

PART C Exit Ticket

6 What's the difference between an exact duplicate and a duplicate id with different values?

WEEKLY QUIZ
Planned for Fri Nov 20

About 15 minutes

Week 3 Quiz

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Concepts and code (1 point each)

1. A left join of 600 scored rows to examinees returns 600 rows. What does that suggest?

   A duplicate keys
   B one match per id
   C missing ids

2. Which item is hardest on Form A?

3. Long format has how many rows for 600 people × 30 items?

4. R function for long → wide

5. Number of distinct raw spellings of program in messy_enrollment

6. How many exact duplicate rows did messy_enrollment have?

7. Write a one-line check that every program value is Day, Evening, or Online (any language).

LESSON 2.9 · PRACTICE
Planned for Mon Nov 30

Unit 2: Data Wrangling

Lesson 2.9: SQL for Analysts I: SELECT, WHERE, GROUP BY

Name _____ Date _____

TODAY'S GOAL

I can write SQL queries that filter and summarize a table, and run them from SAS, Python, and R.

Standards: SQL · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.9 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 SQL clause that names the table

2 SQL function that counts rows

PART B Practice

3 Change the query to prior_gpa >= 3.5. How many examinees qualify in total?

4 Write a query: number of examinees by region, only regions with more than 150 people.

PART C Exit Ticket

5 Why can't you use WHERE COUNT(*) > 150?

LESSON 2.10 · PRACTICE
Planned for Tue Dec 1

Unit 2: Data Wrangling

Lesson 2.10: SQL for Analysts II: Joins and Subqueries

Name _____ Date _____

TODAY'S GOAL

I can join tables and use a subquery in SQL.

Standards: SQL · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.10 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 A query inside a query

2 Keyword for a table alias

PART B Practice

3 Write a query that counts all examinees above the mean, without grouping.

4 Write a query for mean total by gender, using a join.

PART C Exit Ticket

5 Translate the subquery example into a plain-English sentence.

LESSON 2.11 · PRACTICE
Planned for Wed Dec 2

Unit 2: Data Wrangling

Lesson 2.11: Don't Repeat Yourself: Functions and Macros

Name _____ Date _____

TODAY'S GOAL

I can write a reusable function (Python, R) or macro (SAS) for a task I'd otherwise copy and paste.

Standards: Programming · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.11 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 A value passed into a function

2 SAS statement that ends a macro definition

PART B Practice

3 Write a function that returns the mean total for one program, given scored_info and a program name. Test it on Online.

4 Use a loop or comprehension to get the proportion correct for all 30 items, and report the lowest.

PART C Exit Ticket

5 Name a task at work you'd turn into a function or macro.

LESSON 2.12 · PRACTICE
Planned for Thu Dec 3

Unit 2: Data Wrangling

Lesson 2.12: Putting It Together: A Scoring Pipeline

Name _____ Date _____

TODAY'S GOAL

I can build an end-to-end script that imports, scores, computes subscores, joins demographics, and exports a clean file.

Standards: Wrangling · Psychometrics · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.12 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 A score on part of a test

2 Items in each domain

PART B Practice

3 Build the pipeline in the language you use least. Record the subscore means.

4 Add a validation check that reasoning + content + application equals total for every row.

PART C Exit Ticket

5 Why is a subscore based on 10 items less reliable than the total?

WEEKLY QUIZ
Planned for Fri Dec 4

About 15 minutes

Week 4 Quiz

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Concepts and code (1 point each)

1 Which clause filters groups after aggregation?

A WHERE
B HAVING
C ORDER BY

2 Order: SELECT, WHERE, FROM, GROUP BY (as written)

3 What does %mend do?

4 Mean Application subscore

5 Output of p_correct(scored, "i05")

6 Write a SQL query for mean hours_prep by gender.

7 Apply It: Name one step of your team's scoring or reporting process you could script.

LESSON 2.13 · PRACTICE
Planned for Mon Dec 7

Unit 2: Data Wrangling

Lesson 2.13: Working with Dates

Name _____ Date _____

TODAY'S GOAL

I can compute differences between dates and summarize by month.

Standards: Wrangling · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.13 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 SAS counts days from...

2 pandas: extract the month

PART B Practice

3 Latest enrollment date

4 Enrollments in September

5 Compute days_before for every row and report the mean.

PART C Exit Ticket

6 Why separate a date's stored value from its format?

LESSON 2.14 · PRACTICE
Planned for Tue Dec 8

Unit 2: Data Wrangling

Lesson 2.14: Working with Text

Name _____ Date _____

TODAY'S GOAL

I can extract, change, and search text values, including simple patterns.

Standards: Wrangling · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.14 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 A text value

2 Part of a string

PART B Practice

3 Write a pattern that checks every id is E followed by exactly four digits. Do all 600 pass?

4 Regex for text that starts with a capital letter

PART C Exit Ticket

5 Name one text field at work you could analyze with these tools.

LESSON 2.15 · PRACTICE
Planned for Wed Dec 9

Unit 2: Data Wrangling

Lesson 2.15: Codebooks and Data Documentation

Name _____ Date _____

TODAY'S GOAL

I can write a codebook that explains every variable in a data set.

Standards: Documentation · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.15 first.
- [ ] Do every part on these pages: Part A Warm-Up, Part B Practice, Part C Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1 A readable description of a variable

2 Text shown in place of a code

PART B Practice

3 Write codebook rows for content, application, and pass (from Lesson 2.2).

PART C Exit Ticket

4 Who benefits from a codebook besides you?

LESSON 2.16 · PRACTICE
Planned for Thu Dec 10

Unit 2: Data Wrangling

Lesson 2.16: Wrangling Project: Enrollment Meets Scores

Name _____ Date _____

TODAY'S GOAL

I can clean, de-duplicate, join, validate, document, and summarize a messy file from start to finish.

Standards: Project · 30 min

WHAT TO DO

- [ ] Read Textbook Lesson 2.16 first.
- [ ] Do every part on these pages: Part A Project.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Project

1 Paste or summarize your pipeline's key steps.

2 Mean hours_prep for Active enrollees (after joining examinees)

3 Write your two-sentence summary for a program director.

UNIT REVIEW
Planned for Thu Dec 10 (homework) or as needed

About 30 minutes

Unit 2 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Review

1. Filter vs select

2. Left vs inner join

3. Wide vs long

4. Three cleaning fixes for messy text categories

5. WHERE vs HAVING

6. Why write functions?

UNIT EXAM
Planned for Fri Dec 11

About 30 minutes

Unit 2 Exam

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 15 points.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A

Part A · Concepts (1 point each)

1 Which is a ratio-level variable?

A program
B hours_prep
C a 1–5 agreement rating

2 Which pandas method combines tables on a key?

A melt
B merge
C pivot

3 Which statement about missing data is true?

A 3% missing per item always means 3% incomplete rows
B listwise deletion can drop many more rows than any single item's missing rate
C missing values are always random

4 What does validate="one_to_one" check?

5 SAS: what does proc sort … nodupkey; by _all_; do?

PART B Part B · Read the code (1 point each)

6 What does this compute? `long.groupby("item")["score"].mean()`

7 What does this return for Form A? `SELECT COUNT(*) FROM examinees WHERE program = 'Day';`

8 Result of `input(compress("$328.00", "$,"), ?? best12.)`

PART C Part C · Write code (2 points each)

9 Score item i01 (key B) as 0/1 in any language, treating omits as 0.

10 Mean total by program, given a table scored_info with total and program.

PART D Part D · Apply (3 points)

11 Describe a reproducible pipeline for a score file you work with: the steps, two validation checks, and what you'd put in the codebook.

ANSWERS AND CODE SOLUTIONS

Answer Key

Try every problem before you look. Code solutions show one good way; yours may differ and still be right if the output matches.

Unit 1: Your Data Toolkit

Lesson 1.1: Your Toolkit and the Analysis Workflow

#	Question	Answer
1	What does IDE stand for?	integrated development environment
2	Why does reproducible work matter?	Others (or future you) can rerun it and get the same results.
3	Check off each step and write where you'll work: (1) SAS Studio opened; (2) R and RStudio installed or Posit Cloud account; (3) Python with Jupyter ready; (4) data pack downloaded into a folder named data.	Answers vary. All four steps completed.
4	Run the Hello, data example in all three languages. Record what each one printed.	`Each prints its hello message and 35.34 (SAS: x=35.34 in the log; R: [1] 35.34).`
5	Which workflow step do you feel most and least confident about at work? Why?	Answers vary.
6	Write the recurring work task you'll rebuild in code this year.	Answers vary.

Lesson 1.2: Reading Data into SAS, Python, and R

#	Question	Answer
1	CSV stands for…	comma-separated values
2	SAS's name for a data frame	data set

#	Question	Answer
3	Import examinees.csv in the language of your choice. How many rows and columns does it have?	`examinees <- read_csv("data/examinees.csv"); dim(examinees)` `# 600 rows, 6 columns (id, gender, region, program, hours_prep, prior_gpa)`
4	What are the gender, region, and program of examinee E1002?	M, East, Evening
5	How many blank (omitted) responses does exam.csv have on item i01?	7

6	Write the code in a second language to print the first 3 rows of examinees.	`proc print data=work.examinees(obs=3); run;`
		`print(examinees.head(3))`
		`head(examinees, 3)`

7	List the three checks you should do after every import.	dimensions (rows/columns), first rows, column types

Lesson 1.3: Variables, Data Types, and Levels of Measurement

#	Question	Answer
1	Categories with no order	nominal
2	Equal spacing and a true zero	ratio
3	prior_gpa	interval (treated as a continuous number)
4	left_next_year (0/1)	nominal (binary)
5	tenure_years	ratio
6	dept	nominal
7	intent_stay (1–10 rating)	ordinal
8	rank in class	ordinal
9	Load survey.csv and show each column's type. Which columns are numbers but really ordinal?	`survey = pd.read_csv("data/survey.csv"); print(survey.dtypes)` `# e1-e4, w1-w4, s1-s4 (1-5 agreement) and intent_stay are ordinal`

#	Question	Answer
10	Is a mean of 3.02 on one Likert item meaningful? Give one argument each way.	For: with many responses it summarizes the center reasonably. Against: the item is ordinal, so spacing between options may not be equal; report the median or distribution too.

Lesson 1.4: Reproducible Work Habits

#	Question	Answer
1	Folder for the original data	data/raw
2	A system that records every change to files	version control (e.g., Git)
3	Log line “WARNING: Variable X is uninitialized.” What should you do?	Check the code: X was used before it was created (often a typo).
4	Better file name: final_v2_NEW.sas or 2026-11-05_scoring.sas?	2026-11-05_scoring.sas
5	Why rerun a script from the top before sharing?	to make sure results don't depend on steps run out of order or leftover objects

6	Write code to save only the Evening examinees to evening_examinees.csv. How many rows?	`evening = examinees[examinees["program"] == "Evening"]`
		`evening.to_csv("output/evening_examinees.csv", index=False)`
		`# 163 rows`

7	Describe the folder structure you created for your work project.	Answers vary.

Unit 1 Review — Answer Key

Use this on Thursday evening or Friday morning before the exam.

Answers

#	Question	Answer
1	Six workflow steps	question, import, clean, explore, model, communicate
2	Level of measurement for program	nominal
3	Three checks after import	dimensions, first rows, column types
4	Where to fix a data error	in code, not in the raw file

Week 1 Quiz — Answer Key

Answers

#	Question	Answer
1	Which comes right after importing data in the workflow? A. modeling B. cleaning and tidying C. communicating	B. cleaning and tidying
2	A 1–5 agreement item is measured at which level? A. nominal B. ordinal C. ratio	B. ordinal
3	Hours of study is measured at which level? A. ordinal B. interval C. ratio	C. ratio
4	In tidy data, what does each row represent?	one observation
5	What should you never edit by hand?	raw data
6	What does this print? `x <- 17.67 * 2` `x`	[1] 35.34
7	In pandas, what does exam.shape return for exam.csv?	(600, 32)
8	What does proc contents show?	the variables, their types, lengths, and other attributes of a data set
9	Name one recurring task at work, the level of measurement of its main variable, and one summary that fits it.	Answers vary; level and summary must match (e.g., ordinal survey item → median and percent agree).

Scoring

Score	What it means / next step
9–10 (85%+)	Great start.
7–8	Review the lessons you missed.
Below 7	Redo the Code It examples before Unit 2.

Unit 2: Data Wrangling

Lesson 2.1: Selecting Columns and Filtering Rows

#	Question	Answer
1	Keep only certain columns =	select
2	SAS statement to keep rows	WHERE

#	Question	Answer
3	How many examinees are in the South region and the Evening program?	`examinees |> filter(region == "South", program == "Evening") |> nrow()` `# 49`
4	How many examinees have prior_gpa of 3.5 or higher?	`(examinees["prior_gpa"] >= 3.5).sum()  # 81`
5	Keep Day or Online examinees (use “in a list”). How many rows?	`data work.day_online; set work.examinees; where program in ("Day", "Online"); run;` `/* 437 observations */`
6	Why check the row count after every filter?	To confirm the condition did what you meant; a wrong condition (or a typo in a category) silently returns the wrong rows.

Lesson 2.2: Creating Variables: Scoring a Test

#	Question	Answer
1	1 for a match, 0 otherwise is called...	dichotomous scoring
2	Key for item i01	B
3	What proportion of examinees answered i01 correctly?	0.415
4	What proportion answered i05 correctly?	0.96
5	Score the test and report the mean, minimum, and maximum total.	`scored["total"].describe()  # mean 17.67, min 3, max 29`

6	Create pass = 1 if total >= 20, else 0. How many passed?	`scored["pass"] = (scored["total"] >= 20).astype(int);`
		`scored["pass"].sum()  # 237`
		237 of 600 pass.

7	Explain in plain words what the scoring code does.	For each item, it compares each person's answer to the key, records 1 for a match and 0 otherwise, then adds the 30 item scores into a total.

Lesson 2.3: Sorting, Counting, and Grouped Summaries

#	Question	Answer
1	Count of rows in each category	frequency table
2	SAS procedure for counts	PROC FREQ
3	How many Day examinees are female (F)?	150
4	Count examinees by region. Which region is largest?	`examinees["region"].value_counts()  # South 186, East 169, West 129, North 116`
5	Sort examinees by hours_prep, largest first. What is the maximum?	`proc sort data=work.examinees out=work.sorted; by descending hours_prep; run;  /* 58 */`
6	Describe split–apply–combine in your own words with a work example.	Answers vary.

Lesson 2.4: Missing Data: Finding and Handling It

#	Question	Answer
1	pandas: count missing per column	df.isna().sum()
2	R: rows with no missing values	complete.cases()
3	How many respondents are missing at least one item?	264
4	Mean of e1 using all available responses	3.017 (777 responses)

5	Compute each person's mean on e1–e4, using answered items only. Then count people who answered fewer than 3 of the 4.	`eng = ["e1", "e2", "e3", "e4"]`
		`survey["eng_mean"] = survey[eng].mean(axis=1)  # skips NaN`
		`(survey[eng].notna().sum(axis=1) < 3).sum()  # 3 people`
		The rule "at least 3 of 4 answered" is common; record it in your codebook.

6	Why can 3% missing per item turn into 33% incomplete rows?	Missing values fall on different items for different people, so the chance a row is missing something grows with the number of items.

Week 2 Quiz — Answer Key

Answers

#	Question	Answer
1	Which keeps rows? A. select B. filter C. mutate	B. filter
2	pandas operator for “and” between conditions	&
3	What does this return? `(examinees["program"] == "Online").sum()`	154
4	How is an omitted response scored in the Unit 2 scoring code?	0
5	E1002's total score	22
6	Which SAS procedure counts categories? A. PROC MEANS B. PROC FREQ C. PROC SORT	B. PROC FREQ
7	How many survey respondents have complete data on all 12 items?	536
8	Why might listwise deletion bias a survey result?	If people with missing data differ systematically (e.g., less engaged), dropping them changes the estimate.

Lesson 2.5: Joining Tables

#	Question	Answer
1	The column used to match rows	key
2	Join that keeps every row of the main table	left join
3	Mean total for females (F)	17.75
4	Mean total for the South region	18.08
5	What is the correlation between hours_prep and total? (Join first.)	`scored_info["hours_prep"].corr(scored_info["total"])  # 0.449`
6	What would you suspect if a left join turned 600 rows into 612?	Duplicate keys in the right table: some ids matched more than one row.
7	Write the question you'd ask before choosing left vs inner join.	Do I need to keep every row of my main table even if there's no match?

Lesson 2.6: Reshaping Data: Wide and Long

#	Question	Answer
1	One row per person, one column per item	wide
2	R function for wide → long	pivot_longer()
3	Which three items are easiest?	i05 (0.96), i29 (0.918), i26 (0.875)

#	Question	Answer
4	Add each item's domain to the long data (join with exam_key) and compute mean proportion correct by domain.	`long |> left_join(key, by = "item") |> group_by(domain) |> summarise(p = mean(score))` `# Reasoning 0.638, Content 0.590, Application 0.539`
5	Give one analysis that's easier in wide format and one that's easier in long.	Wide: total scores, correlations between items. Long: statistics for every item by group, plots by item.

Lesson 2.7: Cleaning Messy Data

#	Question	Answer
1	Remove spaces at the ends of text	strip / trim
2	Convert, turning failures into missing	coerce
3	How many distinct raw spellings does program have?	8
4	Status counts after cleaning	Active 48, Withdrawn 15
5	What is the mean cleaned fee (all 63 rows, ignoring missing)?	`messy["fee_c"].mean().round(2)  # 280.28`
6	In which month did most enrollments occur?	August
7	Why fix data in code rather than in Excel?	The fix is documented, repeatable, and can be reapplied when new data arrive.

Lesson 2.8: Duplicates and Validation Checks

#	Question	Answer
1	pandas: remove exact duplicates	drop_duplicates()
2	R: stop if a condition fails	stopifnot()
3	After de-duplicating, how many Active enrollments?	45
4	Withdrawn by program (Day, Evening, Online)	9, 4, 2
5	Write a validation check that every total score in your scored data is between 0 and 30.	`assert scored["total"].between(0, 30).all()` `stopifnot(all(scored$total >= 0 & scored$total <= 30))`
6	What's the difference between an exact duplicate and a duplicate id with different values?	Exact duplicates are copies and can be dropped; duplicate ids with different values are conflicting records that need investigation.

Week 3 Quiz — Answer Key

Answers

#	Question	Answer
1	A left join of 600 scored rows to examinees returns 600 rows. What does that suggest? A. duplicate keys B. one match per id C. missing ids	B. one match per id
2	Which item is hardest on Form A?	i17
3	Long format has how many rows for 600 people × 30 items?	18,000
4	R function for long → wide	pivot_wider()
5	Number of distinct raw spellings of program in messy_enrollment	8
6	How many exact duplicate rows did messy_enrollment have?	3
7	Write a one-line check that every program value is Day, Evening, or Online (any language).	`assert dedup["program_c"].isin(["Day", "Evening", "Online"]).all()`

Lesson 2.9: SQL for Analysts I: SELECT, WHERE, GROUP BY

#	Question	Answer
1	SQL clause that names the table	FROM
2	SQL function that counts rows	COUNT(*)
3	Change the query to prior_gpa >= 3.5. How many examinees qualify in total?	`SELECT COUNT(*) FROM examinees WHERE prior_gpa >= 3.5;  -- 81`
4	Write a query: number of examinees by region, only regions with more than 150 people.	`SELECT region, COUNT(*) AS n FROM examinees GROUP BY region HAVING COUNT(*) > 150;` `-- South 186, East 169`
5	Why can't you use WHERE COUNT(*) > 150?	WHERE runs before grouping, so counts don't exist yet; use HAVING.

Lesson 2.10: SQL for Analysts II: Joins and Subqueries

#	Question	Answer
1	A query inside a query	subquery
2	Keyword for a table alias	AS

#	Question	Answer
3	Write a query that counts all examinees above the mean, without grouping.	`SELECT COUNT(*) FROM scored WHERE total > (SELECT AVG(total) FROM scored);` `-- 315 (147 + 91 + 77)`

4	Write a query for mean total by gender, using a join.	`SELECT e.gender, ROUND(AVG(s.total), 2) AS mean_total`
		`FROM scored s JOIN examinees e ON s.id = e.id GROUP BY e.gender;`
		`-- F 17.75, M 17.57`

5	Translate the subquery example into a plain-English sentence.	For each program, count examinees whose total is above the overall average total.

Lesson 2.11: Don't Repeat Yourself: Functions and Macros

#	Question	Answer
1	A value passed into a function	argument
2	SAS statement that ends a macro definition	%mend
3	Write a function that returns the mean total for one program, given scored_info and a program name. Test it on Online.	`mean_total <- function(df, prog) mean(df$total[df$program == prog])` `mean_total(scored_info, "Online")  # 17.47`
4	Use a loop or comprehension to get the proportion correct for all 30 items, and report the lowest.	`ps = {it: p_correct(scored, it) for it in items}; min(ps, key=ps.get)  # i17, 0.307`
5	Name a task at work you'd turn into a function or macro.	Answers vary.

Lesson 2.12: Putting It Together: A Scoring Pipeline

#	Question	Answer
1	A score on part of a test	subscore
2	Items in each domain	10
3	Build the pipeline in the language you use least. Record the subscore means.	`Reasoning 6.38, Content 5.90, Application 5.39, Total 17.67`
4	Add a validation check that reasoning + content + application equals total for every row.	`assert (final[["reasoning", "content", "application"]].sum(axis=1) == final["total"]).all()`

#	Question	Answer
5	Why is a subscore based on 10 items less reliable than the total?	Fewer items means more measurement error relative to the true score (you'll quantify this in Unit 6).

Week 4 Quiz — Answer Key

Answers

#	Question	Answer
1	Which clause filters groups after aggregation? A. WHERE B. HAVING C. ORDER BY	B. HAVING
2	Order: SELECT, WHERE, FROM, GROUP BY (as written)	SELECT, FROM, WHERE, GROUP BY
3	What does %mend do?	ends a SAS macro definition
4	Mean Application subscore	5.39
5	Output of p_correct(scored, "i05")	0.96
6	Write a SQL query for mean hours_prep by gender.	`SELECT gender, AVG(hours_prep) FROM examinees GROUP BY gender;`
7	Apply It: Name one step of your team's scoring or reporting process you could script.	Answers vary.

Lesson 2.13: Working with Dates

#	Question	Answer
1	SAS counts days from…	January 1, 1960
2	pandas: extract the month	.dt.month
3	Latest enrollment date	2026-09-11
4	Enrollments in September	20
5	Compute days_before for every row and report the mean.	`dedup["days_before"].mean()  # 39.0 days (range 20 to 59)`
6	Why separate a date's stored value from its format?	So calculations use the number while people see a readable date in any style.

Lesson 2.14: Working with Text

#	Question	Answer
1	A text value	string
2	Part of a string	substring
3	Write a pattern that checks every id is E followed by exactly four digits. Do all 600 pass?	`examinees["id"].str.fullmatch(r"E\d{4}").all()  # True`
4	Regex for text that starts with a capital letter	^[A-Z]
5	Name one text field at work you could analyze with these tools.	Answers vary.

Lesson 2.15: Codebooks and Data Documentation

#	Question	Answer
1	A readable description of a variable	label
2	Text shown in place of a code	value label / format

3	Write codebook rows for content, application, and pass (from Lesson 2.2).	`content: Content subscore, integer, 0-10, sum of s11-s20`
		`application: Application subscore, integer, 0-10, sum of s21-s30`
		`pass: Passed (total >= 20), 0/1, 1 = passed`

4	Who benefits from a codebook besides you?	Colleagues, reviewers, future analysts, auditors, and you in six months.

Lesson 2.16: Wrangling Project: Enrollment Meets Scores

#	Question	Answer
1	Paste or summarize your pipeline's key steps.	`Clean → de-duplicate → left join with scores → validate → summarize by status → document.`
2	Mean hours_prep for Active enrollees (after joining examinees)	19.76
3	Write your two-sentence summary for a program director.	Answers vary; must state the comparison clearly without jargon.

Unit 2 Review — Answer Key

Use this on Thursday evening or Friday morning before the exam.

Answers

#	Question	Answer
1	Filter vs select	filter keeps rows; select keeps columns
2	Left vs inner join	left keeps all rows of the left table; inner keeps only matches
3	Wide vs long	wide: one column per item; long: one row per person-item
4	Three cleaning fixes for messy text categories	trim spaces, standardize case, map variants (on-line → online)
5	WHERE vs HAVING	WHERE filters rows before grouping; HAVING filters groups after
6	Why write functions?	reuse logic; fix bugs once

Unit 2 Exam — Answer Key

Answers

#	Question	Answer
1	Which is a ratio-level variable? A. program B. hours_prep C. a 1–5 agreement rating	B. hours_prep
2	Which pandas method combines tables on a key? A. melt B. merge C. pivot	B. merge
3	Which statement about missing data is true? A. 3% missing per item always means 3% incomplete rows B. listwise deletion can drop many more rows than any single item's missing rate C. missing values are always random	B. listwise deletion can drop many more rows than any single item's missing rate
4	What does validate="one_to_one" check?	that the key is unique in both tables
5	SAS: what does proc sort … nodupkey; by _all_; do?	removes exact duplicate rows
6	What does this compute? `long.groupby("item")["score"].mean()`	proportion correct (p-value) for each item
7	What does this return for Form A? `SELECT COUNT(*) FROM examinees WHERE program = 'Day';`	283
8	Result of `input(compress("$328.00", "$,"), ?? best12.)`	328

#	Question	Answer

9	Score item i01 (key B) as 0/1 in any language, treating omits as 0.	`s01 = (resp = "B");  /* SAS */`
		`exam["i01"].eq("B").astype(int)  # Python`
		`as.integer(!is.na(exam$i01) & exam$i01 == "B")  # R`

10	Mean total by program, given a table scored_info with total and program.	`scored_info.groupby("program")["total"].mean()`
11	Describe a reproducible pipeline for a score file you work with: the steps, two validation checks, and what you'd put in the codebook.	Rubric: ordered steps from raw to output (1); two sensible validation checks (1); codebook contents named (1).

Scoring

Score	What it means / next step
13–15 (85%+)	Strong. Move on with confidence.
11–12	Solid. Redo the missed items' lessons.
Below 11	Revisit this unit's Code It examples before moving on.

Data Science & Psychometrics

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A year of professional learning in statistics, data science, and psychometrics, built for a working professional: about 30 minutes a day, Monday through Friday. Every lesson pairs a short reading with worked examples, a suggested video, and hands-on code in SAS, Python, and R, using realistic assessment and survey data.

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INSIDE THIS YEAR

01 Your Data Toolkit

02 Data Wrangling

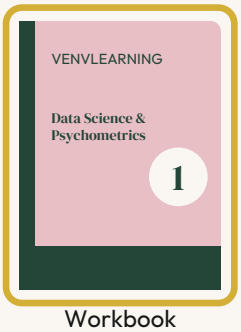
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- Practice and coding exercises for every lesson
- A quiz every Friday
- Unit exams
- Answer key with code solutions

THE COMPLETE SET



Textbook



Workbook

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